AI Question Bank Answers

MODULE 1

1. Discuss the history of Artificial Intelligence.
2. Why knowledge representation is necessary in AI systems? Give AI systems in which knowledge is important?
3. Define: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality, (e) logical reasoning.
4. There are well-known classes of problems that are intractably difficult for computers,
   and other classes that are provably undecidable. Does this mean that AI is impossible?
5. Explain the relationship between evolution and one or more of autonomy, intelligence, and learning.
6. Explain the concept of rationality.
7. Let us examine the rationality of various vacuum-cleaner agent functions.
   a. Show that the simple vacuum-cleaner agent function described in Figure 2.3 is indeed rational under the assumptions listed on page
   b. Describe a rational agent function for the case in which each movement costs one
      point. Does the corresponding agent program require internal state?
   c. Discuss possible agent designs for the cases in which clean squares can become
      dirty and the geography of the environment is unknown. Does it make sense for the
      agent to learn from its experience in these cases? If so, what should it learn? If not,
      why not?

8. What are the properties of task environments? Explain.
9. What do you understand by structure of an Agent?
10. What is agent program? Explain four basic kinds of agent programs.

Answers

▼ 1

Maturation of Artificial Intelligence (1943-1952)

- **Year 1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter pits in 1943. They proposed a model of **artificial neurons**.
- **Year 1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called **Hebbian learning**.
- **Year 1950:** The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes **"Computing Machinery and Intelligence"** in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a **Turing test**.

The birth of Artificial Intelligence (1952-1956)

- **Year 1955:** An Allen Newell and Herbert A. Simon created the "first artificial intelligence program"Which was named as **"Logic Theorist"**. This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.
- **Year 1956:** The word "Artificial Intelligence" first adopted by American Computer scientist John McCarthy at the Dartmouth Conference. For the first

time, AI coined as an academic field.

At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

The golden years-Early enthusiasm (1956-1974)

- **Year 1966:** The researchers emphasized developing algorithms which can solve mathematical problems. Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.
- **Year 1972:** The first intelligent humanoid robot was built in Japan which was named as WABOT-1.

The first AI winter (1974-1980)

- The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.
- During AI winters, an interest of publicity on artificial intelligence was decreased.

A boom of AI (1980-1987)

- **Year 1980:** After AI winter duration, AI came back with "Expert System". Expert systems were programmed that emulate the decision-making ability of a human expert.
- In the Year 1980, the first national conference of the American Association of Artificial Intelligence **was held at Stanford University**.

The second AI winter (1987-1993)

- The duration between the years 1987 to 1993 was the second AI Winter duration.

- Again Investors and government stopped in funding for AI research as due to high cost but not efficient result. The expert system such as XCON was very cost effective.

The emergence of intelligent agents (1993-2011)

- **Year 1997:** In the year 1997, IBM Deep Blue beats world chess champion, Gary Kasparov, and became the first computer to beat a world chess champion.
- **Year 2002:** for the first time, AI entered the home in the form of Roomba, a vacuum cleaner.
- **Year 2006:** AI came in the Business world till the year 2006. Companies like Facebook, Twitter, and Netflix also started using AI.

Deep learning, big data and artificial general intelligence (2011-present)

- **Year 2011:** In the year 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.
- **Year 2012:** Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
- **Year 2014:** In the year 2014, Chatbot "Eugene Goostman" won a competition in the infamous "Turing test."
- **Year 2018:** The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.
- Google has demonstrated an AI program "Duplex" which was a virtual assistant and which had taken hairdresser appointment on call, and lady on other side didn't notice that she was talking with the machine.

▼ 2

Humans are best at understanding, reasoning, and interpreting knowledge. Human knows things, which is knowledge and as per their knowledge they perform various actions in the real world. **But how machines do all these things comes under knowledge representation and reasoning**

- Knowledge representation and reasoning (KR, KRR) is the part of Artificial intelligence which concerned with AI agents thinking and how thinking contributes to intelligent behavior of agents.
- It is responsible for representing information about the real world so that a computer can understand and can utilize this knowledge to solve the complex real world problems such as diagnosis a medical condition or communicating with humans in natural language.
- It is also a way which describes how we can represent knowledge in artificial intelligence. Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

▼ 3

- (a) **Intelligence** refers to the ability of an individual or system to process information, learn from experience, and make decisions or solve problems.
- (b) **Artificial intelligence (AI)** refers to the simulation of human intelligence in machines that are programmed to think and learn like humans.
- (c) An **agent** refers to a system or entity that can perceive its environment and take actions to achieve a goal or accomplish a task.
- (d) **Rationality** refers to the ability of an individual or system to make logical and consistent decisions or take appropriate actions.
- (e) **Logical reasoning** refers to the process of using logical arguments and inference to arrive at a conclusion or decision. It's a method of thinking that involves making inferences based on premises and logical rules.

▼ 4

No, it does not mean that AI is impossible. While there are certain classes of problems that are intractably difficult for computers and other classes that are provably undecidable, there are still many problems that can be solved using AI techniques. Additionally, AI research continues to make progress in developing new techniques and algorithms that can solve increasingly complex problems.

▼ 5

- Evolution and autonomy, intelligence, and learning are related in several ways.
- Firstly, evolution can be seen as a process of learning through trial and error, over time. Through the process of natural selection, organisms that are better suited to their environment have a higher chance of survival and reproduction, and their genetic traits are passed on to future generations. Over time, this process leads to the evolution of increasingly complex and adaptive organisms.
- Secondly, autonomy, intelligence and learning are characteristics that have evolved in many organisms as a way to adapt to their environment. Autonomy refers to the ability of an organism to act independently and make decisions based on its own perception and cognition. Intelligence refers to the ability to process information and make decisions. And learning refers to the ability to improve one's performance based on experience. These characteristics have evolved in many organisms as a way to improve their chances of survival and reproduction.
- Lastly, evolution has also played a role in the development of Artificial Intelligence. AI systems are designed to learn and adapt like living organisms, by using techniques such as machine learning and deep learning which are inspired by the way the brain processes information. These AI systems are then tested and improved over time, similar to the process of natural selection in evolution.
- In conclusion, evolution is a process that has led to the development of autonomy, intelligence, and learning in many organisms, and these characteristics have also been used in the development of AI systems.

▼ 6

rationality refers to the ability to make logical and consistent decisions or take appropriate actions based on evidence and reason. It is a quality that is associated with decision-making, philosophy and psychology, and it can be improved by understanding and addressing the limitations and biases that can affect our ability to be rational.

▼ 7

1. The simple vacuum-cleaner agent function described in Figure 2.3 is a simple rule-based agent that moves from location to location in a grid, cleaning the squares as it goes. The agent function is assumed to have the following properties:

- The agent can sense the state of the square it is currently on (clean or dirty).
- The agent can move to an adjacent square (up, down, left, or right).
- The agent has a fixed set of rules for deciding which square to move to next.

Under these assumptions, the agent function can be considered rational as it is able to achieve its goal of cleaning all the squares in the grid by following a set of rules that are based on the current state of the environment.

1. In the case in which each movement costs one point, a rational agent function would be one that balances the cost of movement with the benefit of cleaning a dirty square. This can be achieved by using a cost function that assigns a higher value to dirty squares that are adjacent to already cleaned squares, and a lower value to dirty squares that are farther away. An agent program that implements this cost function would require internal state to keep track of the squares that have already been cleaned.
2. In the cases in which clean squares can become dirty and the geography of the environment is unknown, it makes sense for the agent to learn from its experience. In this case, the agent should learn the patterns of dirtiness, so it can prioritize cleaning certain areas more often or at certain times. Additionally, the agent should learn the location of the dirtier areas, so it can navigate to them more efficiently. If the environment is large and complex, it may also make sense for the agent to learn and use a map of the environment.

In summary, the rationality of an agent function depends on the assumptions and goals of the problem, and the agent should be able to adapt to the changing environment by learning from its experience.

▼ 8

Properties of task environments

An environment in artificial intelligence is the surrounding of the agent. The agent takes input
from the environment through sensors and delivers the output to the environment through
actuators. There are several types of environments:

- Fully Observable vs Partially Observable
- Deterministic vs Stochastic
- Single-agent vs Multi-agent
- Static vs Dynamic
- Discrete vs Continuous
- Episodic vs Sequential
- Known vs Unknown
- Fully Observable vs Partially Observable
- When an agent sensor is capable to sense or access the complete state of an agent at each point in time, it is said to be a fully observable environment else it is partially observable.
- Maintaining a fully observable environment is easy as there is no need to keep track of
  the history of the surrounding.
- An environment is called unobservable when the agent has no sensors in all environments.
- Examples:
    - o Chess – the board is fully observable, and so are the opponent's moves.
    - o Driving – the environment is partially observable because what's around the corner is not known.
- Deterministic vs Stochastic
- When a uniqueness in the agent's current state completely determines the next state of

the agent, the environment is said to be deterministic.

- □ The stochastic environment is random in nature which is not unique and cannot be
  completely determined by the agent.
- □ Examples:
  - o Chess – there would be only a few possible moves for a coin at the current state
    and these moves can be determined.
    o Self-Driving Cars- the actions of a self-driving car are not unique, it varies time
    to time.
- Single-agent vs Multi-agent
- □ An environment consisting of only one agent is said to be a single-agent environment.
- □ A person left alone in a maze is an example of the single-agent system.

▼ 9

The job of AI is to design an agent program that implements the agent function the mapping
from percepts to actions.
This program will run on some sort of computing device with physical sensors and actuators
called the architecture.
agent = architecture + program
Architecture makes the percepts from the sensors available to the program, runs the program, and
feeds the program's action choices to the actuators as they are generated.
Agent program: use current percept as input from the sensors and return an action to the
actuators.
Agent function: takes the entire percept history

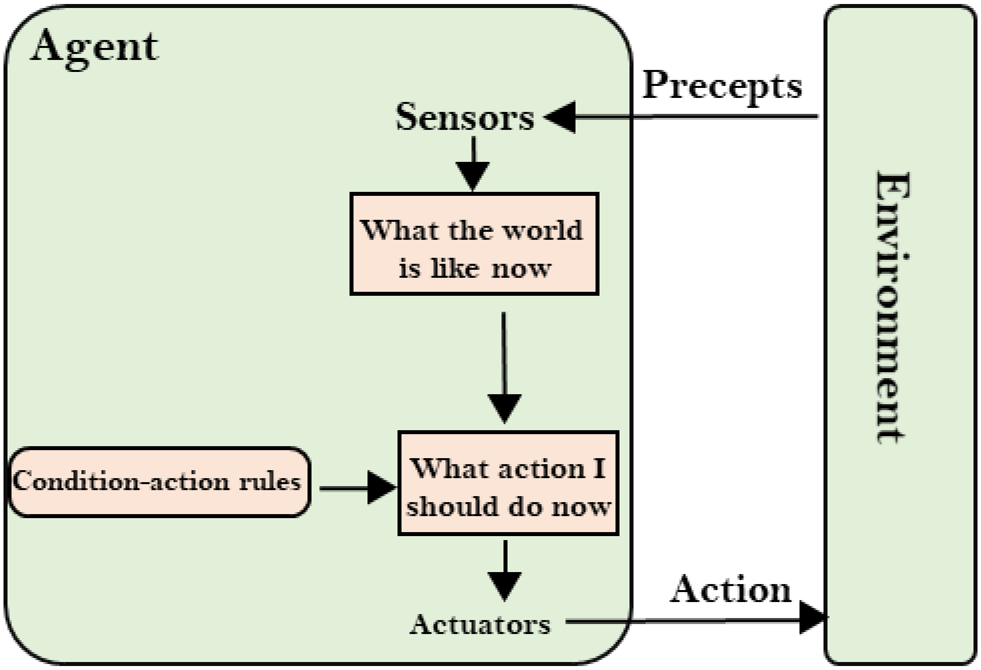
▼ 10

Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time. These are given below:

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

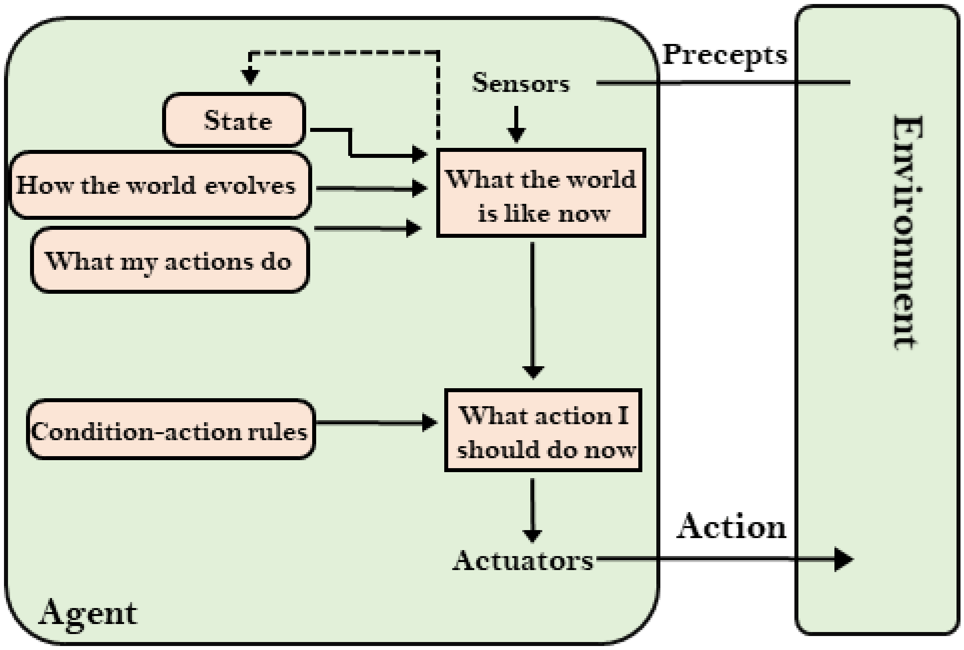
1. Simple Reflex agent:

- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history.
- These agents only succeed in the fully observable environment.
- The Simple reflex agent does not consider any part of percepts history during their decision and action process.
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.
- Problems for the simple reflex agent design approach:
    - They have very limited intelligence
    - They do not have knowledge of non-perceptual parts of the current state
    - Mostly too big to generate and to store.
    - Not adaptive to changes in the environment.



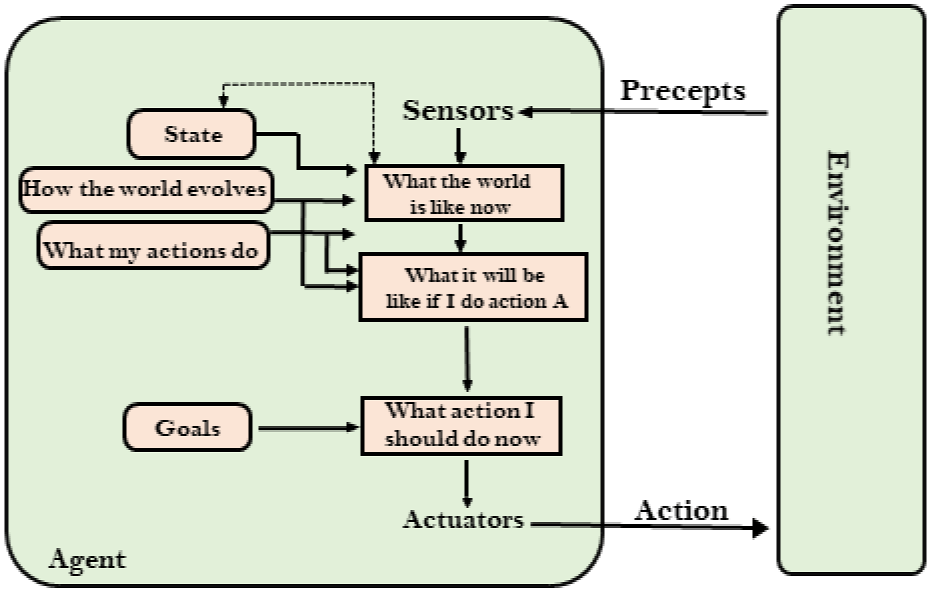
2. Model-based reflex agent

- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
  - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
  - **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:
  1. How the world evolves
  2. How the agent's action affects the world.



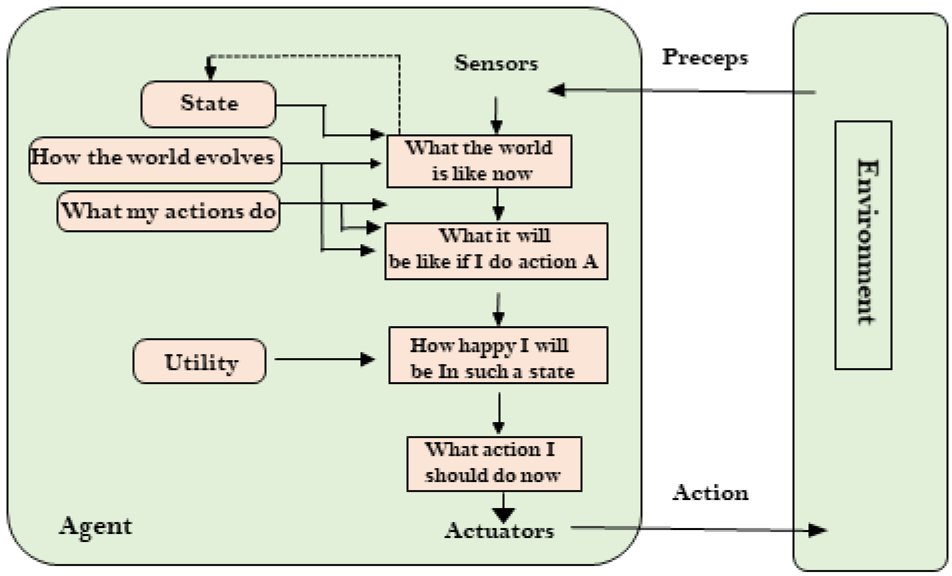
3. Goal-based agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.



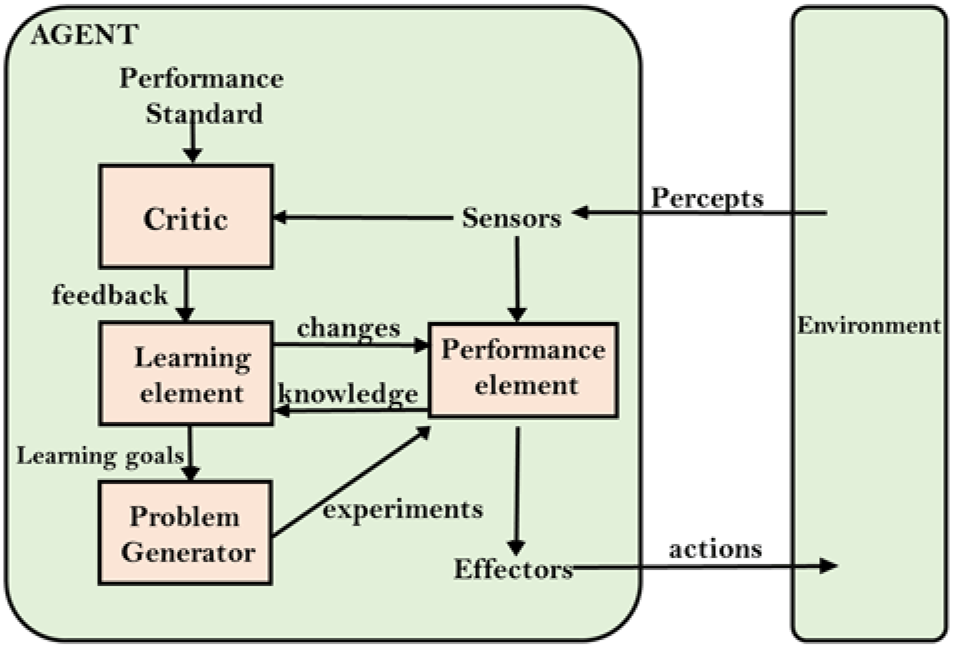
4. Utility-based agents

- These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.



5. Learning Agents

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
    1. **Learning element:** It is responsible for making improvements by learning from environment
    2. **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
    3. **Performance element:** It is responsible for selecting external action
    4. **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.
- Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.


AGENT
Performance Standard
Critic
Sensors
Percepts
feedback
changes
Learning element
knowledge
Performance element
Environment
Learning goals
Problem Generator
experiments
Effectors
actions